

Unseen Seminar Questions 2: Answers

Probability and Statistical Inference

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1. A random variable X has mean 10, standard deviation 2, skewness 1.4, and kurtosis 6. Let $Y = 3 - 2X$. Without finding a density, calculate $\mathbb{E}[Y]$ and $\text{sd}(Y)$, and give the skewness and kurtosis of Y . Interpret the direction of skewness. Explain what the kurtosis suggests about extreme observations relative to a normal benchmark, and state a limitation of this comparison.

Solution. Linearity and the scale rule for standard deviations give

$$\mathbb{E}[Y] = 3 - 2(10) = -17, \quad \text{sd}(Y) = |-2|\text{sd}(X) = 4.$$

Translation and positive rescaling do not change standardized moments. The negative scale reverses the sign of the third standardized moment but not the fourth, so Y has skewness -1.4 and kurtosis 6. It is therefore left-skewed. A normal variable has kurtosis 3, so Y has excess kurtosis 3. This high fourth standardized moment is commonly associated with more influence from extreme observations than under a normal benchmark. However, kurtosis alone does not prove that Y has larger tail probabilities than a normal variable at every threshold. It does not establish that one distribution has heavier tails in every possible sense.

2. Let $X \sim U[-3, 3]$ and $Y = |X|$. Derive the CDF and density of Y , and calculate $\Pr(Y > 2)$. Explain why using only the inverse branch $x = y$ would give the wrong density.

Solution. The support of Y is $[0, 3]$. For $0 \leq y \leq 3$,

$$F_Y(y) = \Pr(-y \leq X \leq y) = \frac{2y}{6} = \frac{y}{3}.$$

Therefore

$$F_Y(y) = \begin{cases} 0, & y < 0, \\ y/3, & 0 \leq y \leq 3, \\ 1, & y > 3, \end{cases} \quad f_Y(y) = \begin{cases} 1/3, & 0 < y < 3, \\ 0, & \text{otherwise.} \end{cases}$$

It follows that $\Pr(Y > 2) = 1 - F_Y(2) = 1/3$. Each interior value y has two inverse images, $x = y$ and $x = -y$. Using only $x = y$ includes only half the probability and would incorrectly produce density $1/6$.

3. The joint probabilities of binary variables X and Y are

	$X = 0$	$X = 1$
$Y = 0$	0.30	0.10
$Y = 1$	0.20	0.40

Calculate $\mathbb{E}[Y | X = 0]$, $\mathbb{E}[Y | X = 1]$, and $\mathbb{E}[Y]$. Verify the law of iterated expectations. Use Bayes' rule to calculate $\Pr(X = 1 | Y = 1)$, identifying the prior, likelihood and normalizing probability. Are X and Y independent? Calculate $\text{Cov}(X, Y)$.

Solution. Both columns have marginal probability 0.5. Since Y is binary,

$$\mathbb{E}[Y | X = 0] = \Pr(Y = 1 | X = 0) = \frac{0.20}{0.50} = 0.4,$$

and

$$\mathbb{E}[Y | X = 1] = \Pr(Y = 1 | X = 1) = \frac{0.40}{0.50} = 0.8.$$

The marginal probability $\Pr(Y = 1)$ is $0.20 + 0.40 = 0.60$, so $\mathbb{E}[Y] = 0.6$. Iterating gives

$$\mathbb{E}[\mathbb{E}(Y | X)] = 0.5(0.4) + 0.5(0.8) = 0.6.$$

For the Bayes calculation, the prior is $\Pr(X = 1) = 0.5$, the likelihood is $\Pr(Y = 1 | X = 1) = 0.8$, and the normalizing probability is $\Pr(Y = 1) = 0.6$. Hence

$$\Pr(X = 1 | Y = 1) = \frac{\Pr(Y = 1 | X = 1) \Pr(X = 1)}{\Pr(Y = 1)} = \frac{(0.8)(0.5)}{0.6} = \frac{2}{3}.$$

The variables are not independent: for example,

$$\Pr(X = 1, Y = 1) = 0.4 \neq (0.5)(0.6) = 0.3.$$

Finally, $\mathbb{E}[X] = 0.5$, $\mathbb{E}[Y] = 0.6$, and $\mathbb{E}[XY] = \Pr(X = 1, Y = 1) = 0.4$, so

$$\text{Cov}(X, Y) = 0.4 - (0.5)(0.6) = 0.1.$$

4. The joint density of (X, Y) is

$$f_{X,Y}(x, y) = 6x, \quad 0 < x < 1, \quad 0 < y < 1 - x,$$

and is zero otherwise. Verify that it integrates to one. Find $f_X(x)$ and $f_{Y|X}(y | x)$, then calculate $\Pr(Y < X)$. Draw the support and shade the event $Y < X$. Use the sketch to explain why the probability integral changes at $x = 1/2$.

Solution. The support is the triangle below $y = 1 - x$ in the positive quadrant. Normalization holds because

$$\int_0^1 \int_0^{1-x} 6x \, dy \, dx = 6 \int_0^1 x(1-x) \, dx = 6 \left(\frac{1}{2} - \frac{1}{3} \right) = 1.$$

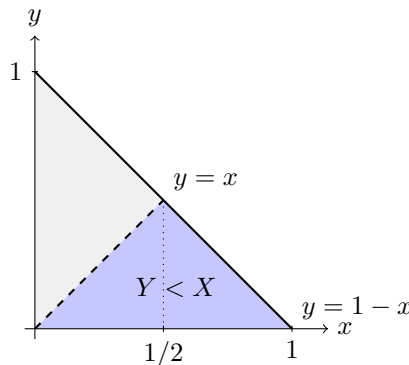
The marginal and conditional densities are

$$f_X(x) = 6x(1-x), \quad 0 < x < 1,$$

and

$$f_{Y|X}(y | x) = \frac{6x}{6x(1-x)} = \frac{1}{1-x}, \quad 0 < y < 1-x.$$

The gray triangle below is the support. The darker triangle is the event $Y < X$ within that support.



For $x < 1/2$, the event $Y < X$ imposes the upper bound $y < x$. For $x > 1/2$, the support bound $y < 1 - x$ is already below x . Therefore

$$\begin{aligned}\Pr(Y < X) &= \int_0^{1/2} \int_0^x 6x \, dy \, dx + \int_{1/2}^1 \int_0^{1-x} 6x \, dy \, dx \\ &= \frac{1}{4} + \frac{1}{2} = \frac{3}{4}.\end{aligned}$$

5. A charity recruits volunteers through social media. It then randomly assigns each volunteer to receive either a new fundraising script ($D = 1$) or the old script ($D = 0$). Let $\bar{Y}_0$ and $\bar{Y}_1$ be the two sample mean donations. Choose $\hat{\alpha}$ and $\hat{\beta}$ so that $\hat{\alpha} + \hat{\beta}D$ equals $\bar{Y}_0$ when $D = 0$ and $\bar{Y}_1$ when $D = 1$. Identify all true statements and briefly correct each false statement.

- (a) $\hat{\beta}$ equals the treated-minus-control difference in sample mean donations.
- (b) $\hat{\alpha}$ equals the mean donation among treated volunteers.
- (c) Random assignment makes the recruited volunteers an independent and identically distributed (iid) random sample of all potential fundraisers.
- (d) If treatment were self-selected instead, a difference in mean donations could reflect selection as well as the script's effect.

Solution. Statements (a) and (d) are true. Setting $D = 0$ gives $\hat{\alpha} = \bar{Y}_0$. Setting $D = 1$ then gives $\hat{\alpha} + \hat{\beta} = \bar{Y}_1$, so $\hat{\beta}$ is $\bar{Y}_1 - \bar{Y}_0$. If treatment were chosen, differences in the kinds of people choosing each script could also generate a difference in means.

Statements (b) and (c) are false. The estimate $\hat{\alpha}$ is the control mean $\bar{Y}_0$; the treated mean is $\hat{\alpha} + \hat{\beta}$. Assignment determines treatment after entry into the study, not who enters. It therefore does not make social-media volunteers a random sample of all potential fundraisers. Applying the result to that wider population requires random sampling or additional assumptions about its average treatment effect.